

Thanh Dang

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EDUCATION

California State Polytechnic University, Pomona

Bachelor of Science, Computer Science

August 2022 – May 2026

GPA: 3.72

- **Honors:** Presidential and Dean's List: 2022-2026

SKILLS

- **Programming Languages:** Python, Java, JavaScript, TypeScript, SQL, HTML, CSS
- **Frameworks:** React, React Native, Next.js, NestJS, Flask, Ray, PyTorch
- **Tools & Technologies:** Firebase, AWS, PostgreSQL, MySQL, Docker, Kafka, Redis, Node.js, Git

EXPERIENCE

AI/ML Researcher & Full Stack Developer, **CPP Enterprises**, Pomona, CA

May 2025 – Present

- Developed GPU-enabled video processing service using Kafka for event streaming and Ray to orchestrate multi-step, asynchronous pipelines, delivering real-time updates via Server-Sent Events and Redis pub/sub
- Implemented NestJS REST API serving multiple features including route design, service-layer logic, and PostgreSQL persistence; deployed on Jetstream2 via CI/CD pipeline and Docker Compose orchestration
- Spearheaded development of audio-based task identification using WhisperX transcription with speaker diarization and Google Gemma LLM few-shot prompting, processing 50+ hours of mobility scooter driving footage
- Trained self-supervised Transformer-based model to quantify upper-body asymmetry from 2D pose sequences, achieving strong separation between stroke and healthy groups using contrastive learning and ranking objectives

Undergraduate Research Assistant, **Cal Poly Pomona**, Pomona, CA

October 2024 – May 2025

- Built Python desktop application for Delta NCSA HPC cluster file management via SFTP, streamlining upload, download, and deletion workflows; presented the project at the CPP RSCA Conference to 20+ attendees
- Utilized YOLO11n-pose to extract normalized upper-body keypoints from clinical session videos, implementing IoU-based person tracking, feeding into downstream stability classification and sway displacement analysis
- Aligned label, motion, and pose data across 40 recording sessions by filtering non-driving segments through temporal segmentation, reducing overall dataset size by 15% while retaining 100% of relevant driving events

Code Coach, **theCoderSchool**, Cerritos, CA

January 2024 – February 2025

- Led coding classes for 30+ elementary students with no prior programming experience, teaching fundamental programming concepts and guiding students to build functional platformers and dialogue-driven games in Scratch
- Coached 5 students one-on-one in front-end web development, advancing from core HTML and CSS layout principles to designing and building interactive web pages with JavaScript, leveraging Replit for live deployment

PROJECTS

TrailTalk

September 2025 – December 2025

- Launched social media web app for park enthusiasts using React and Firebase in Agile team environment, featuring 63 U.S. national parks with dedicated park pages, user reviews, photo sharing, and following features
- Designed scalable class-based NoSQL data model supporting user accounts, messaging, and content management; integrated MapLibre for interactive mapping and geolocation features across 50 states
- Implemented live private messaging system with TypeScript and Cloud Firestore, incorporating notifications and read receipts to enable seamless user-to-user communication with low-latency message delivery

Carbon Closet, **Cal Poly Pomona Hackathon 2025**

April 2025

- Built full-stack gamified e-commerce platform that makes buying, selling, and donating secondhand clothing simple and rewarding through token-based rewards; presented to 8 judges and won 3rd place among 60 teams
- Developed responsive interface with Next.js and Cloud Firestore, supporting live listing updates, real-time search filtering, shopping cart management, user profiles, and gacha bundle purchases with dynamic inventory tracking
- Containerized application with Docker to ensure consistent environments across development and production; deployed to AWS EC2, gaining hands-on experience with cloud hosting and production deployment workflows

Recyclable or Not Bot, **UC Riverside Hackathon 2025**

January 2025

- Created computer vision web app to detect recyclable vs. non-recyclable items, rendering real-time classification, color-coded score to promote eco-friendly waste disposal; awarded best sustainability track among 50+ teams
- Integrated end-to-end pipeline using YOLOv8 deep learning model for object detection, with Flask API managing OpenCV frame capture and delivering inference results to React front-end alongside recycling tips and resources

EXTRACURRICULARS

STEM Advantage Scholar: Scholarship program for first-generation college students in STEM

June 2024 – May 2025

Software Engineer Association: Largest software engineering community at CPP

November 2023 – May 2024